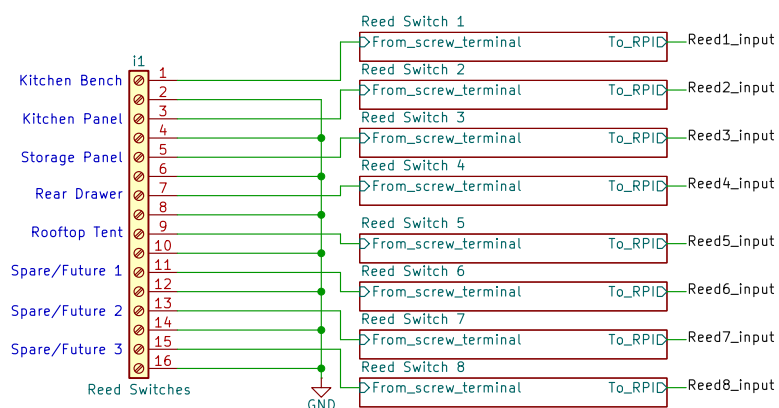
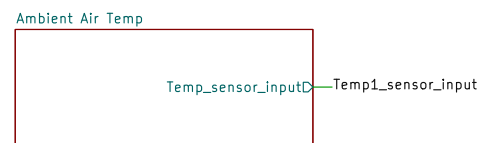


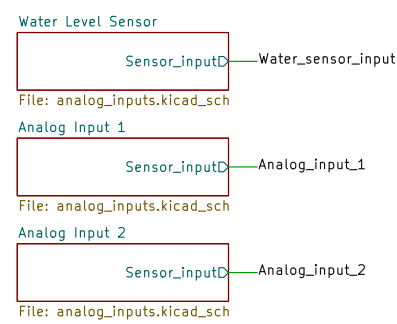
Reed Switches



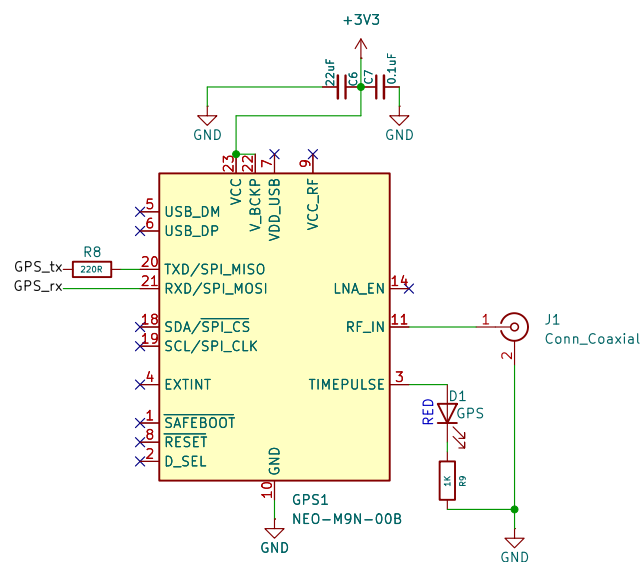
Temp Sensor 1-Wire Buses



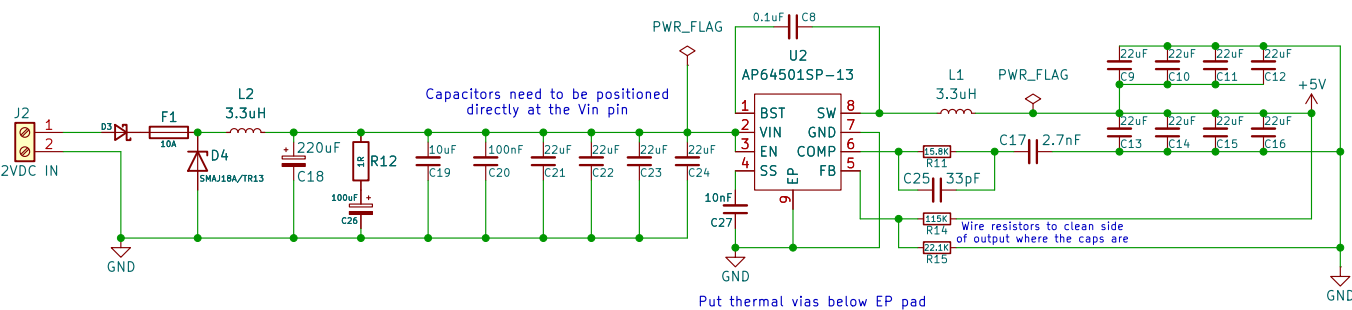
Arduino Analog Inputs



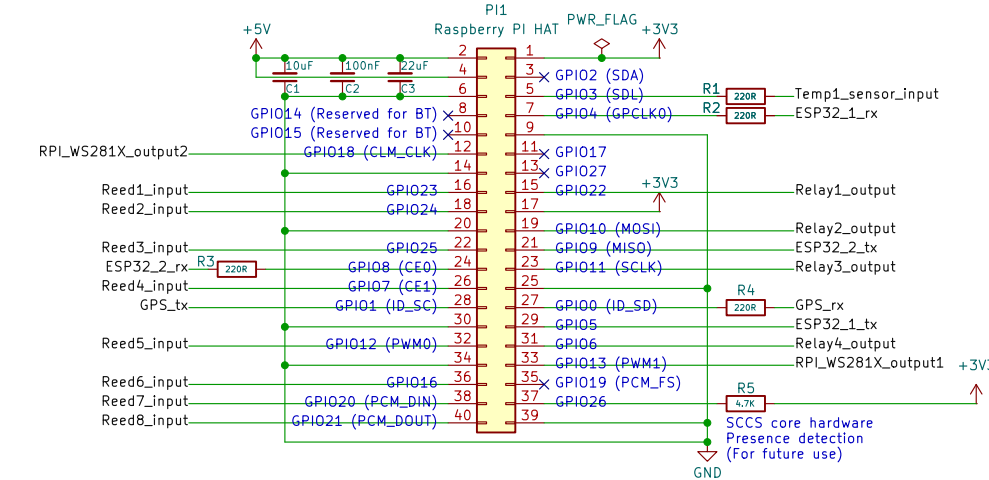
GPS



Power



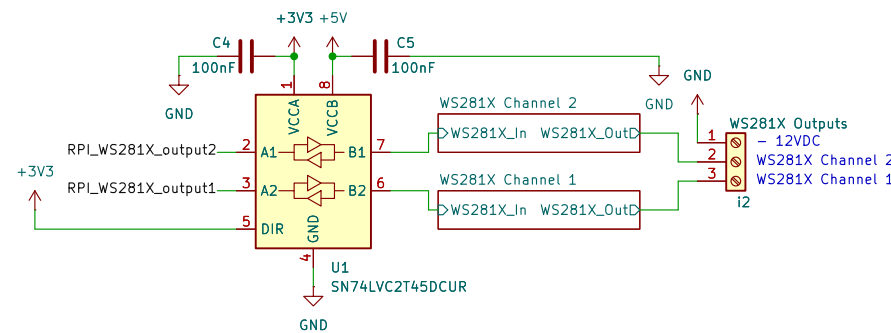
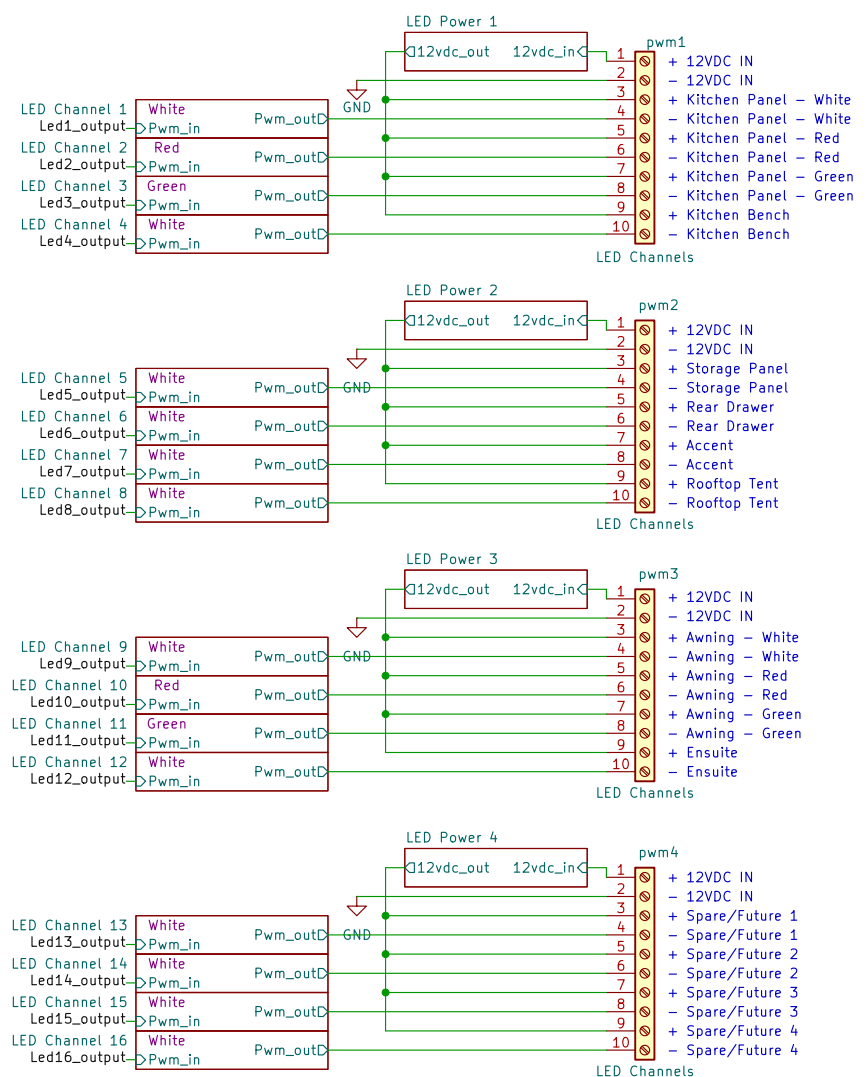
Raspberry Pi



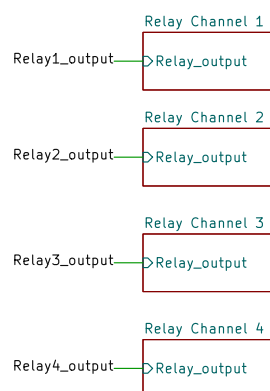
ESP32's



Dimmable Lighting

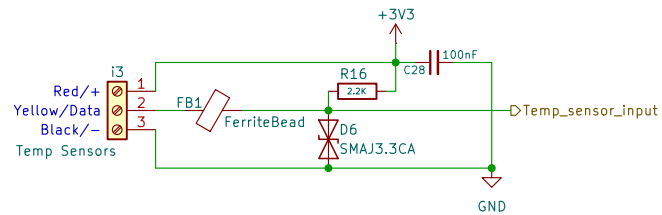


Relay Channels



Notes

- Power circuit provides 5V
- RPI Provides 3.3V
- 1-Wire Bus Notes:
Board has a 2.2K pull-up (not a series resistor) on the data line.
Put a 220R resistor on the data pin of each temp sensor but ensure it's not in series with the bus.
Put a 100nF ceramic cap across VCC and GND at each temp sensor.
- WS281X RPI Notes:
PWM Channel 0 can be routed to GPIO 12 OR GPIO 18
PWM Channel 1 can be routed to GPIO 13 OR GPIO 19
Board already has 33R in series on data – do not add another series resistor at the strip.
Put a 1000uF electrolytic cap between 12V and GND on each LED strip
- UARTS:
UART1 / GPIO14/15 / Reserved for Bluetooth
UART2 / GPIO00/01 / GPS
UART3 / GPIO04/05 / ESP32 No. 1
UART4 / GPIO08/09 / ESP32 No. 2
- 12V LED Strip Notes (not necessary but good practice):
Put a 220uF 25V electrolytic cap across 12V and GND at each LED strip.
Put a 100nF 25V ceramic cap across 12V and GND at each LED strip.



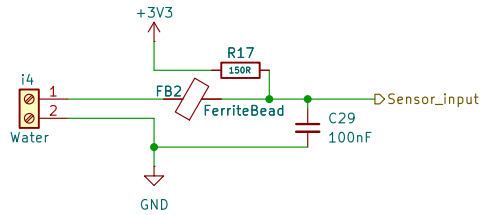
Sheet: /Ambient Air Temp/
File: 1_wire_bus.kicad_sch

Title:

Size: A5
KiCad E.D.A. 10.0.4

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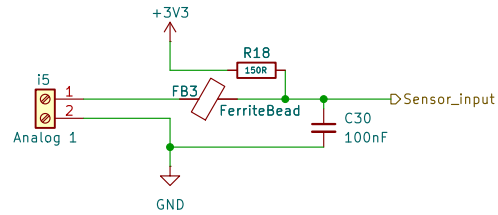
Sheet: /Water Level Sensor/
File: analog_inputs.kicad_sch

Title:

Size: A5
KiCad E.D.A. 10.0.4

Date:

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Id: 3/41



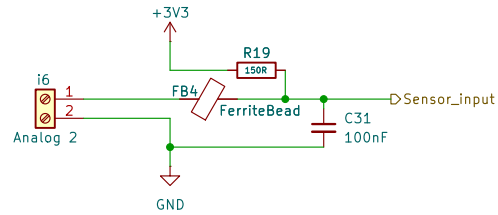
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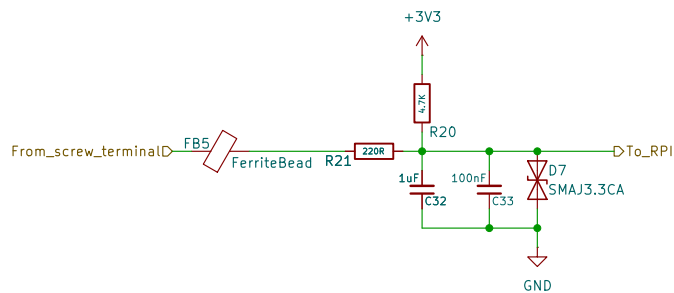
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Title:

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Date:

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Id: 5/41



Sheet: /Reed Switch 1/
File: reed_input.kicad_sch

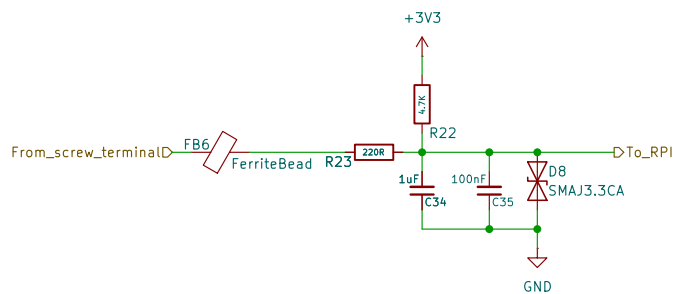
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Date:

Rev:

Id: 6/41



Sheet: /Reed Switch 2/
 File: reed_input.kicad_sch

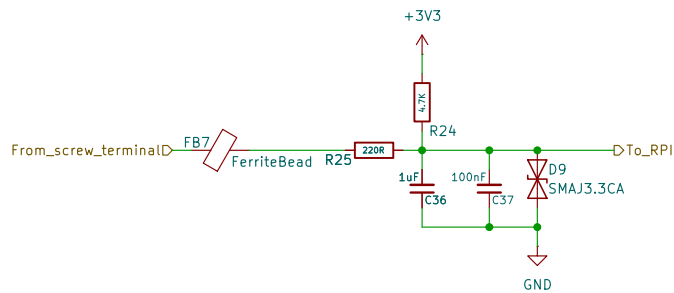
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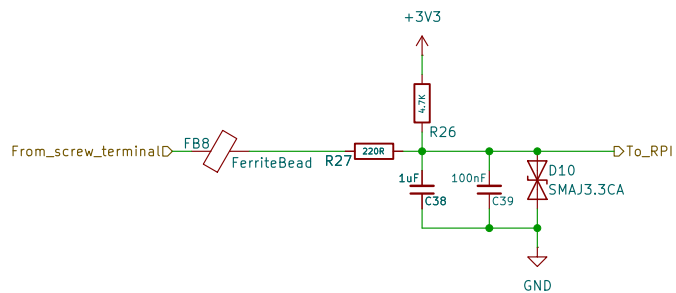
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Rev:
Id: 8/41



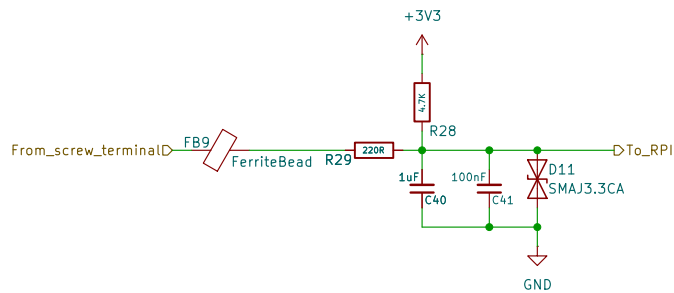
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Date:

Rev:
Id: 9/41



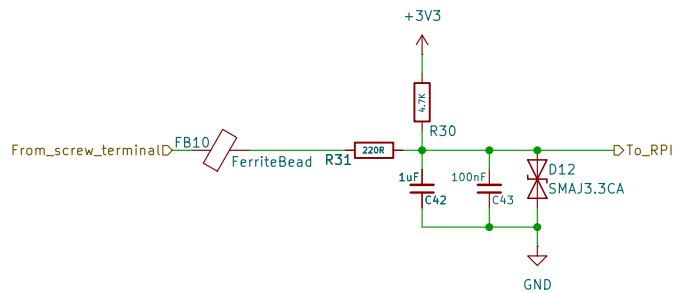
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Date:

Rev:
Id: 10/41



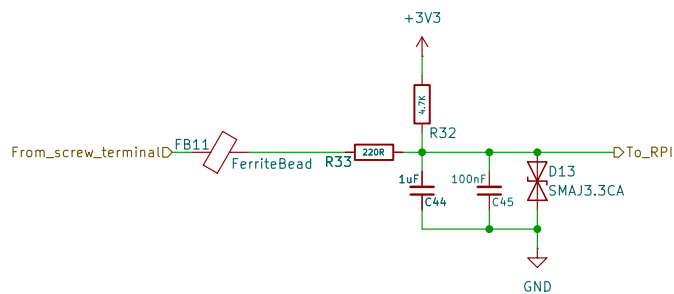
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Date:

Rev:
Id: 11/41



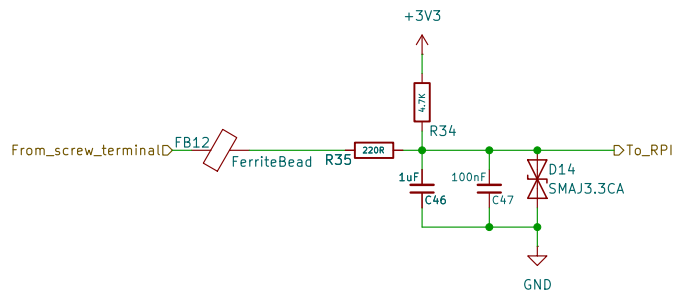
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Date:

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Id: 12/41



Sheet: /Reed Switch 8/
File: reed_input.kicad_sch

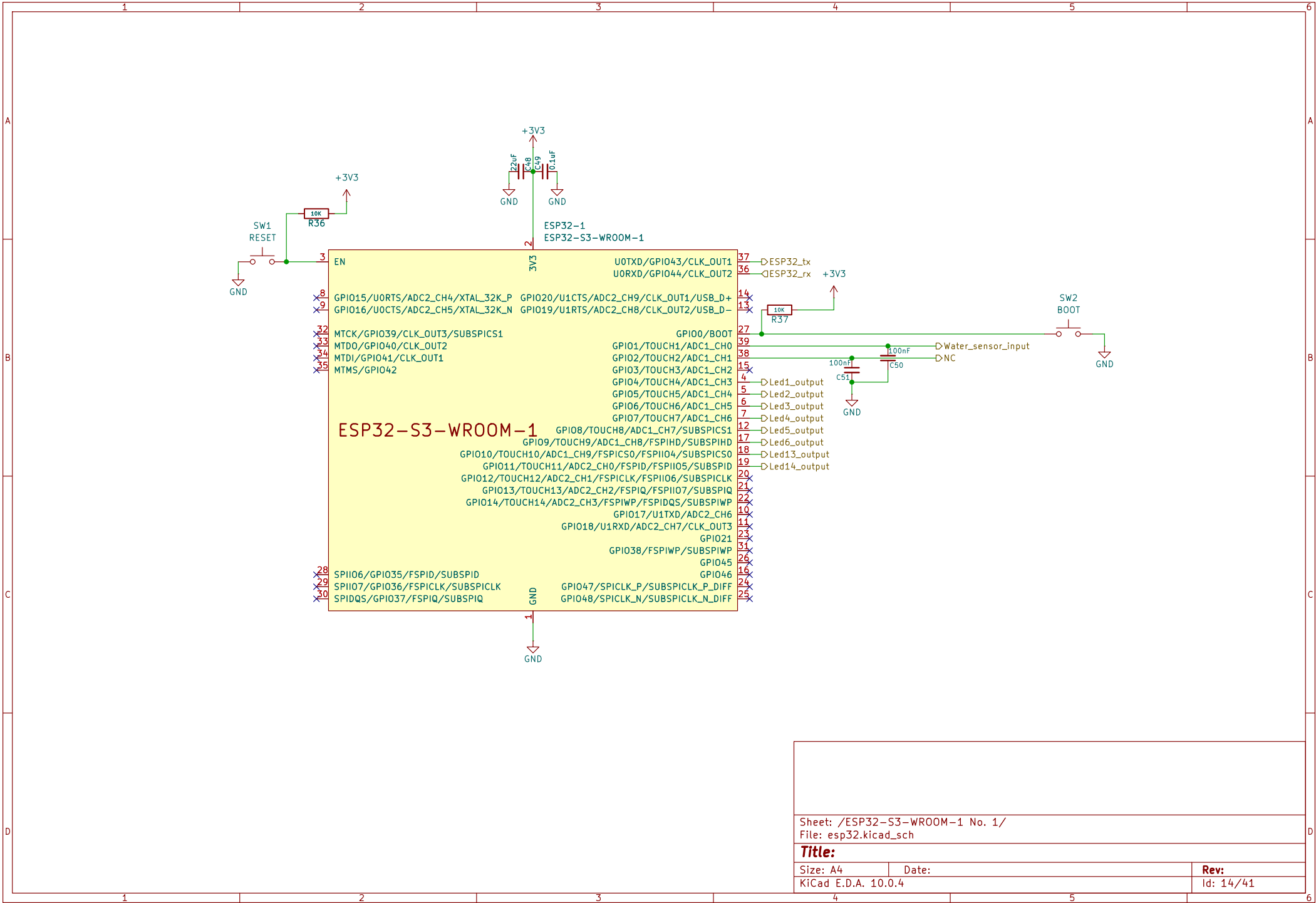
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Date:

Rev:

Id: 13/41



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Size: A4

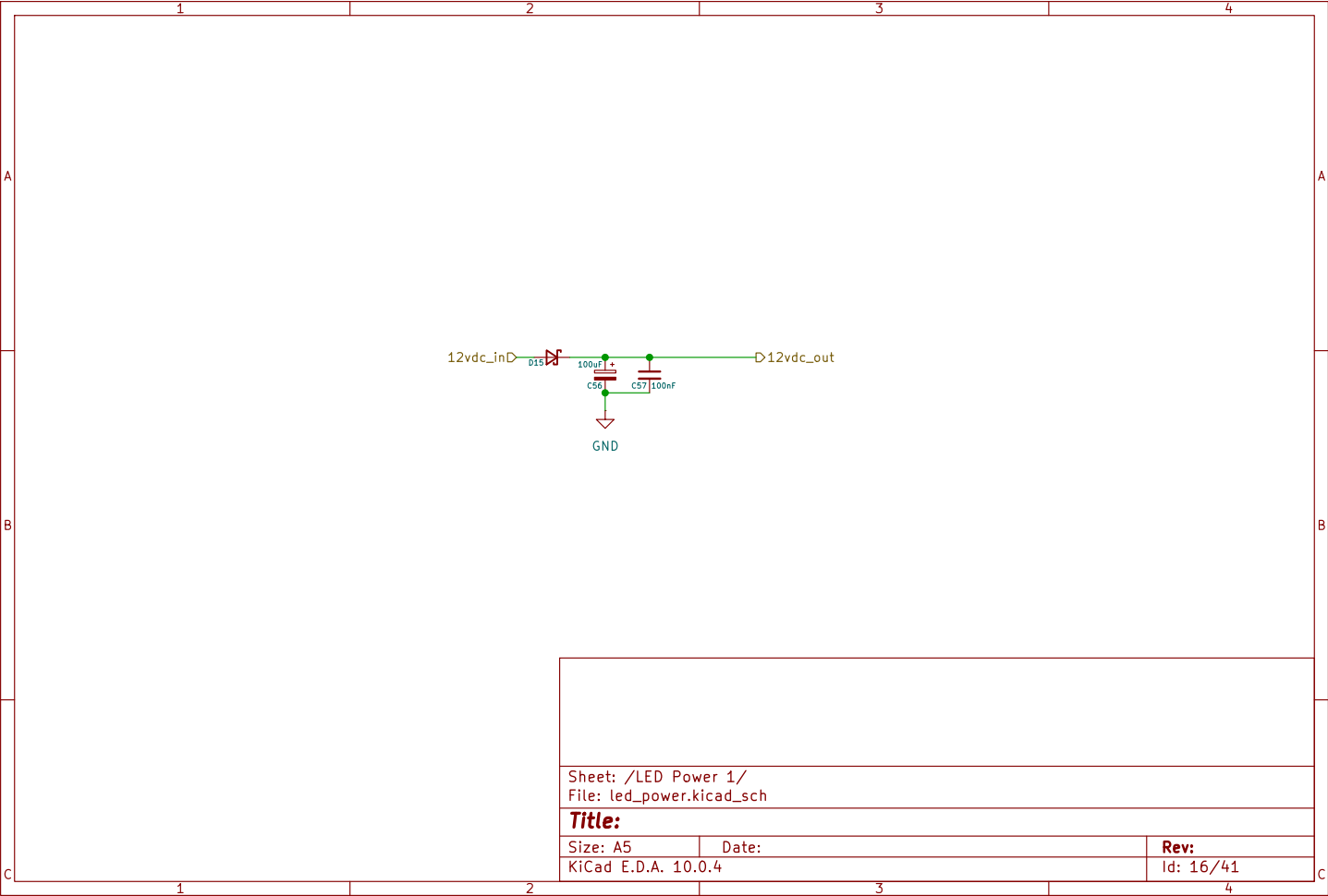
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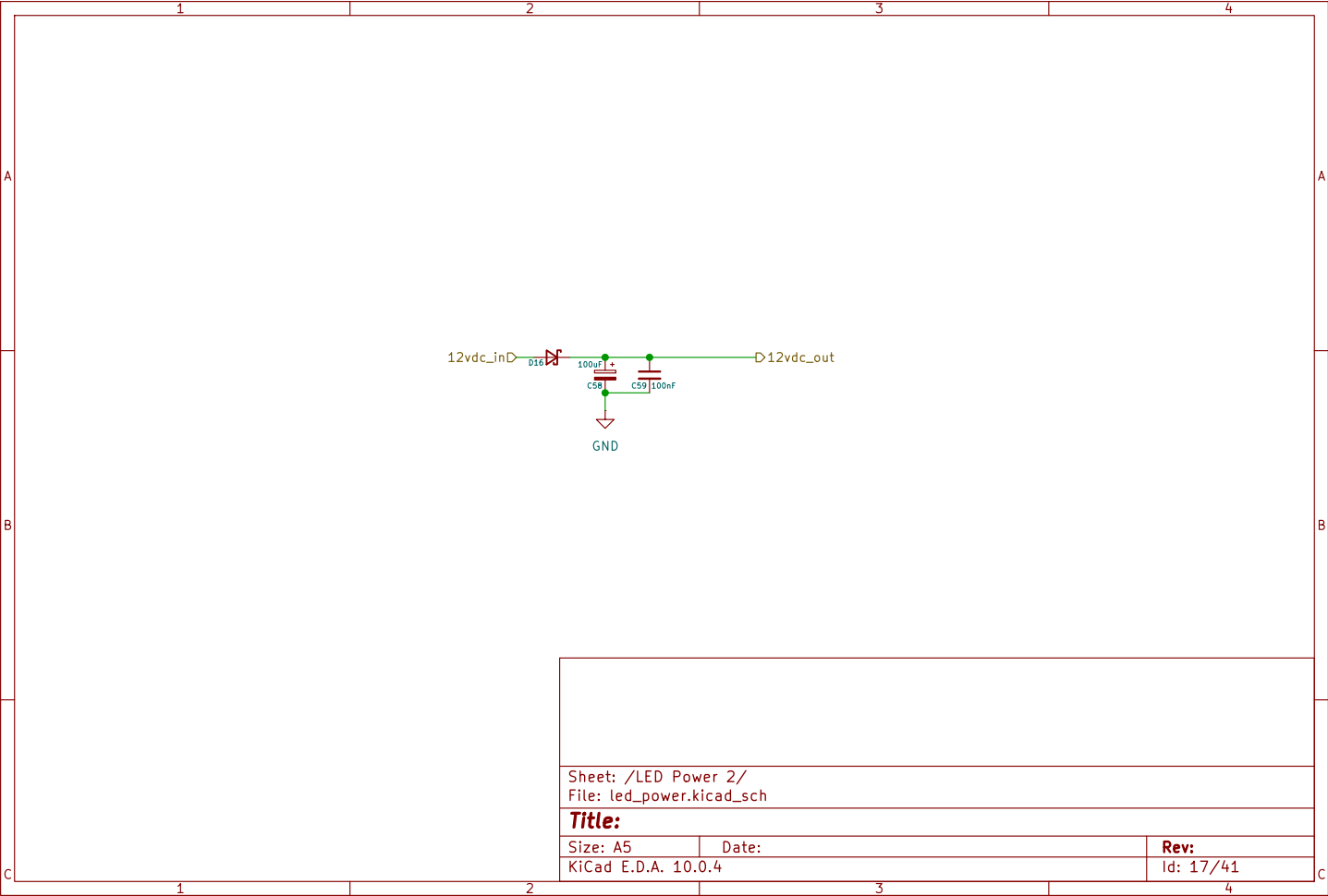
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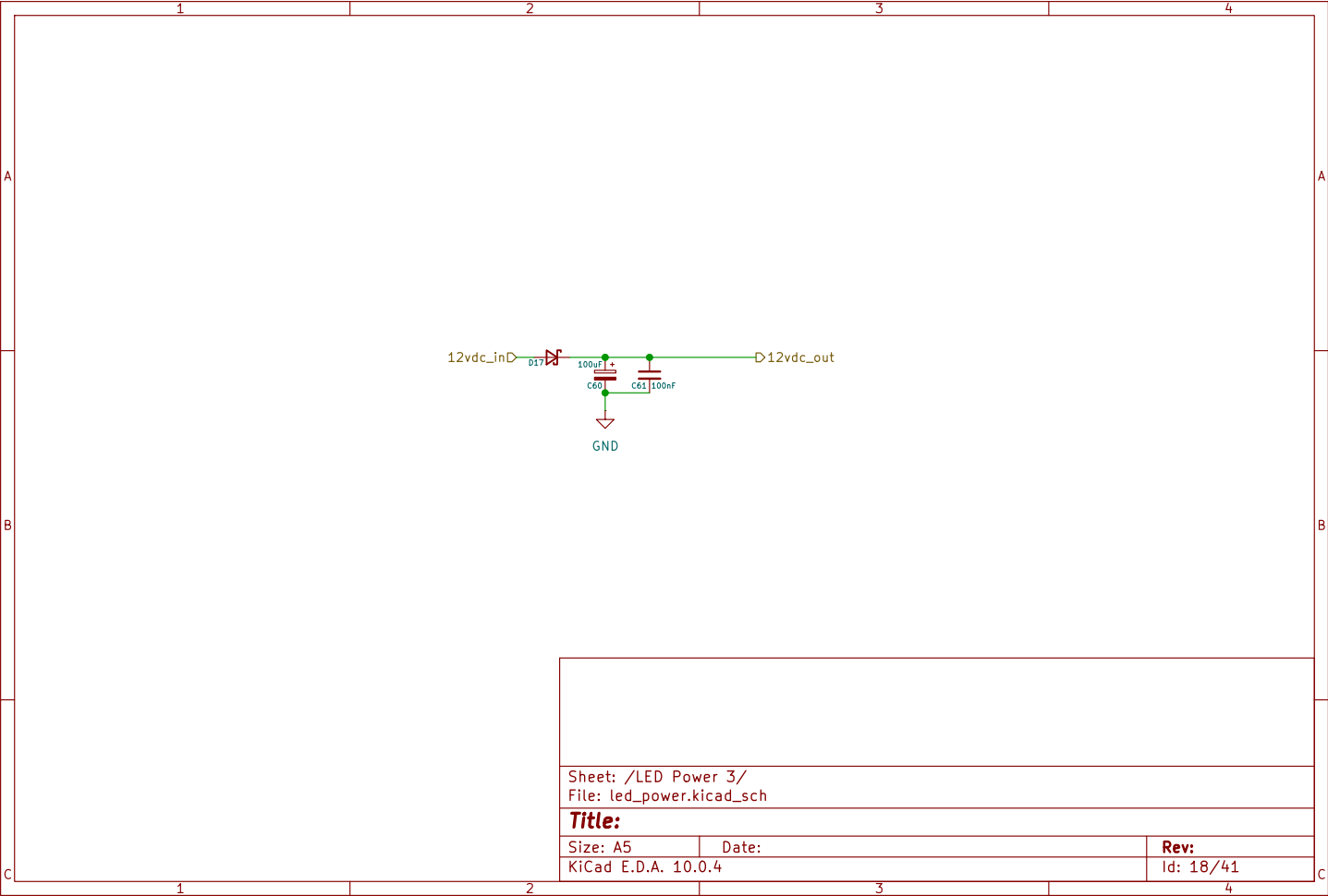
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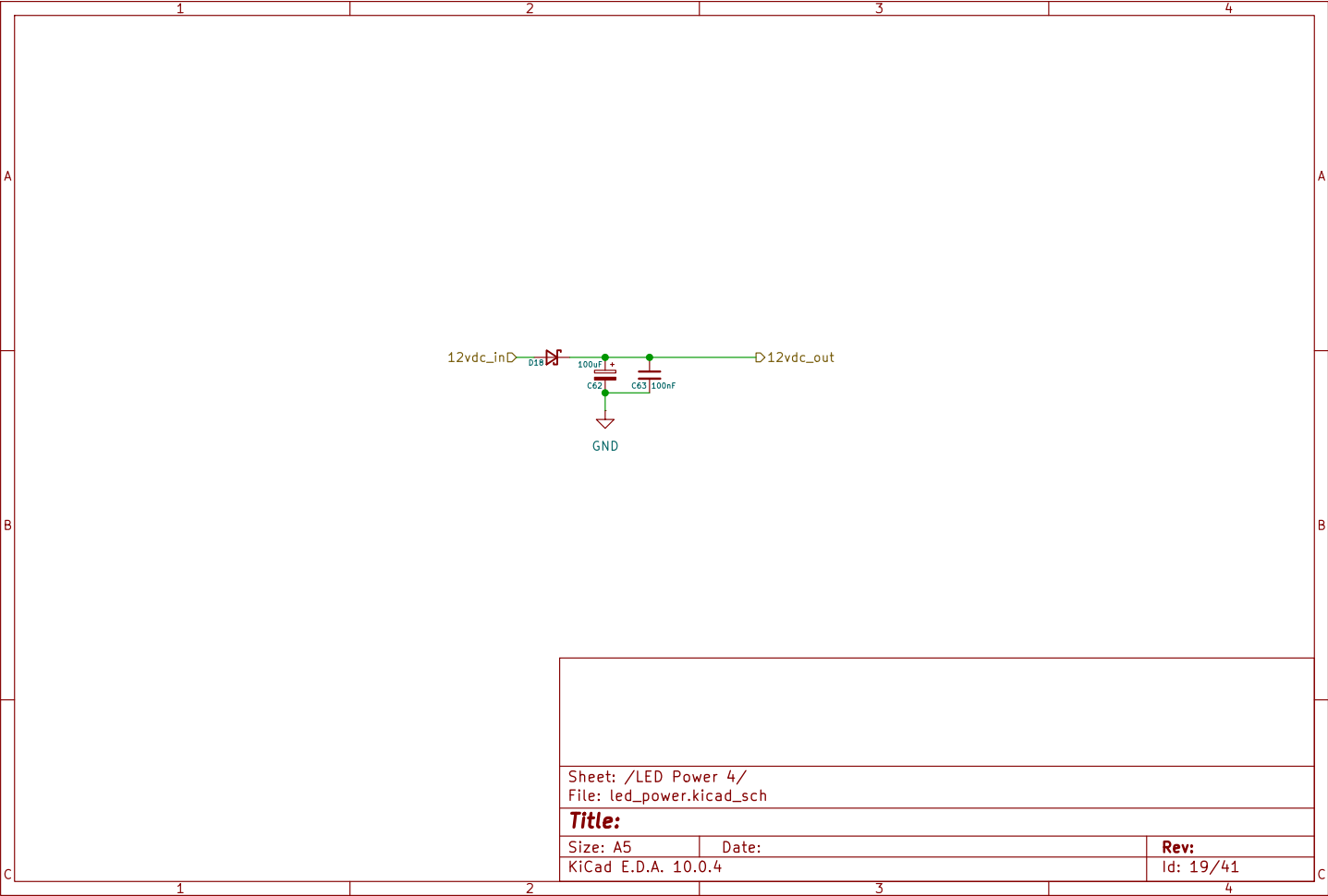
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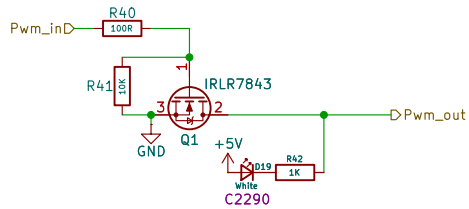




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Size: A5	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 17/41







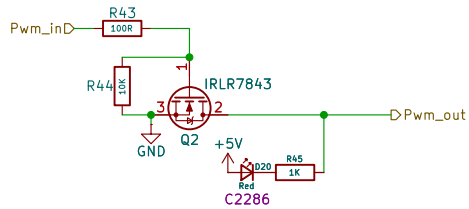
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Date:

Rev:
Id: 20/41



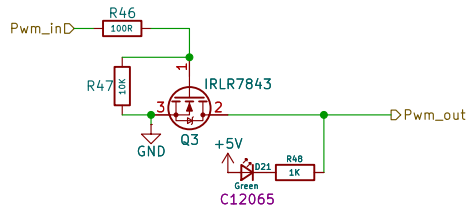
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Rev:
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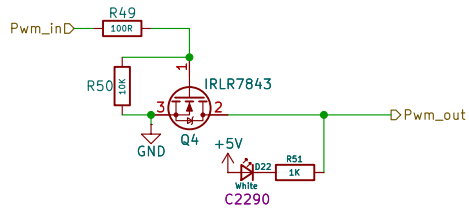
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Date:

Rev:
Id: 23/41



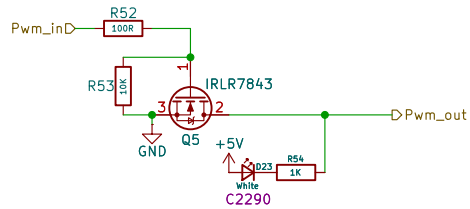
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Id: 24/41



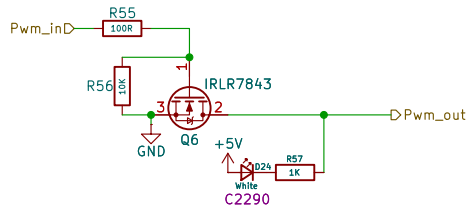
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Rev:
Id: 25/41



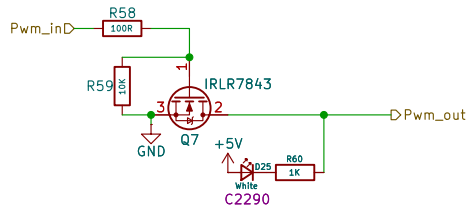
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Date:

Rev:
Id: 26/41



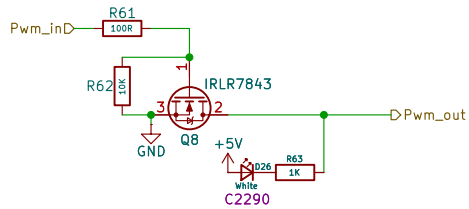
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Rev:
Id: 27/41



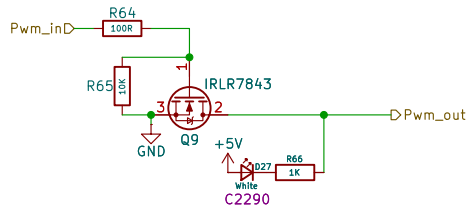
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Date:

Rev:
Id: 28/41



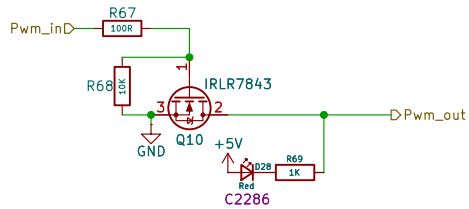
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Rev:
Id: 29/41



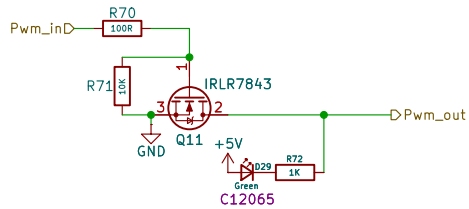
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Date:

Rev:
Id: 30/41



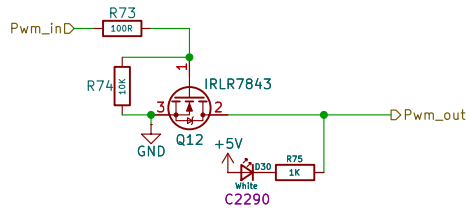
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Date:

Rev:
Id: 31/41



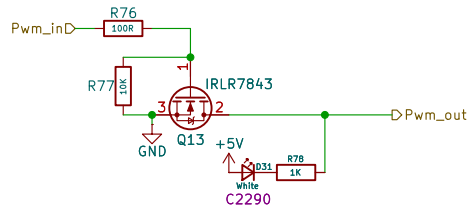
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Date:

Rev:
Id: 32/41



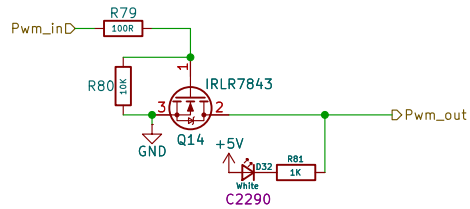
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Date:

Rev:
Id: 33/41



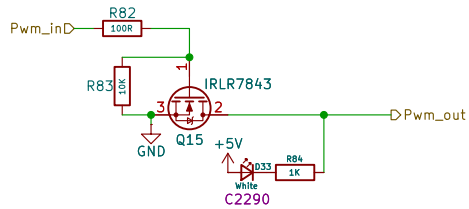
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Date:

Rev:
Id: 34/41



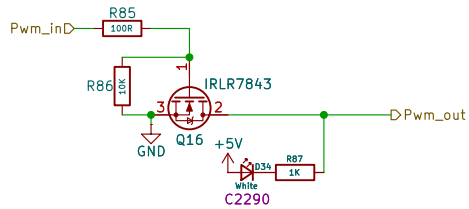
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Date:

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Id: 35/41



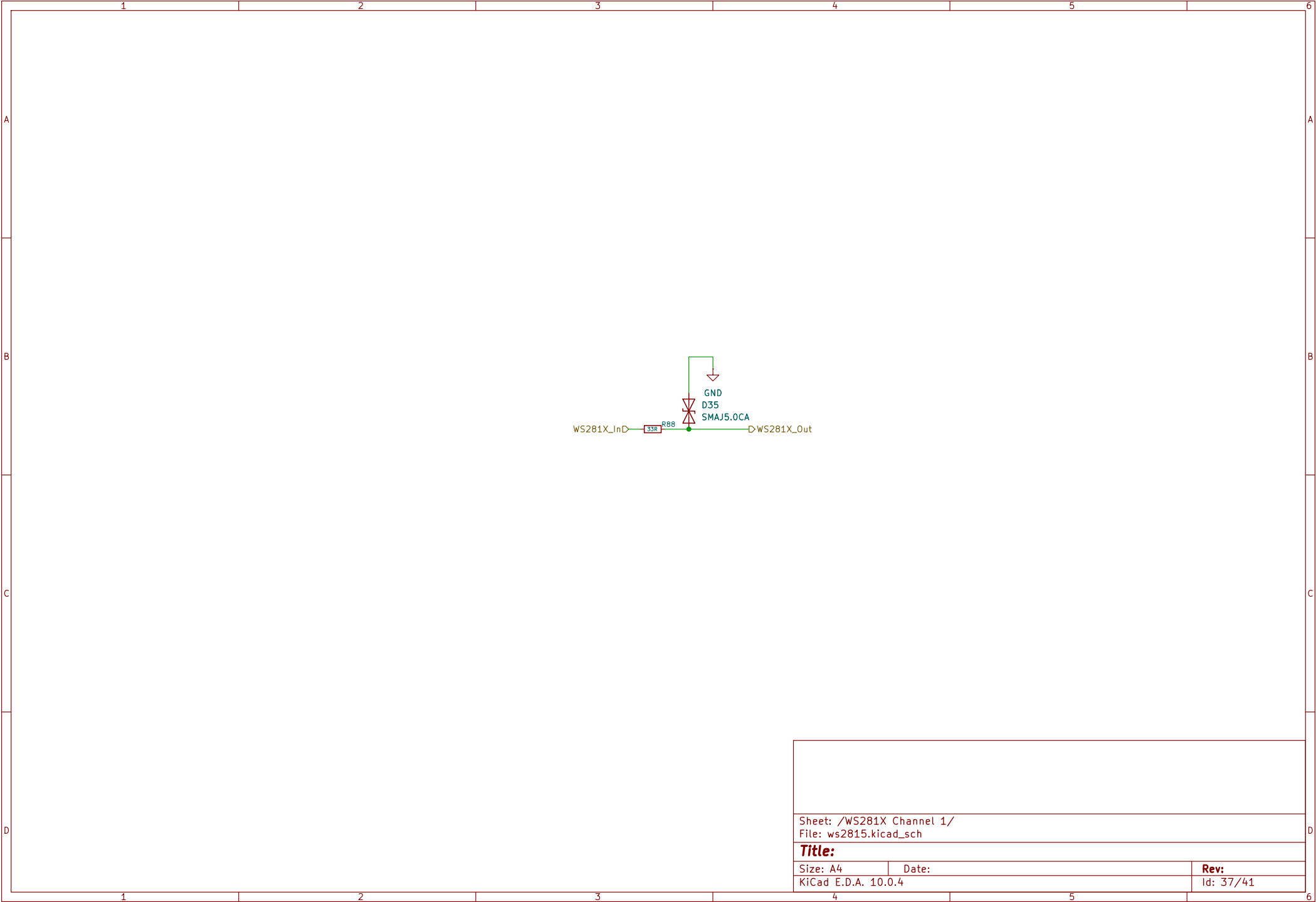
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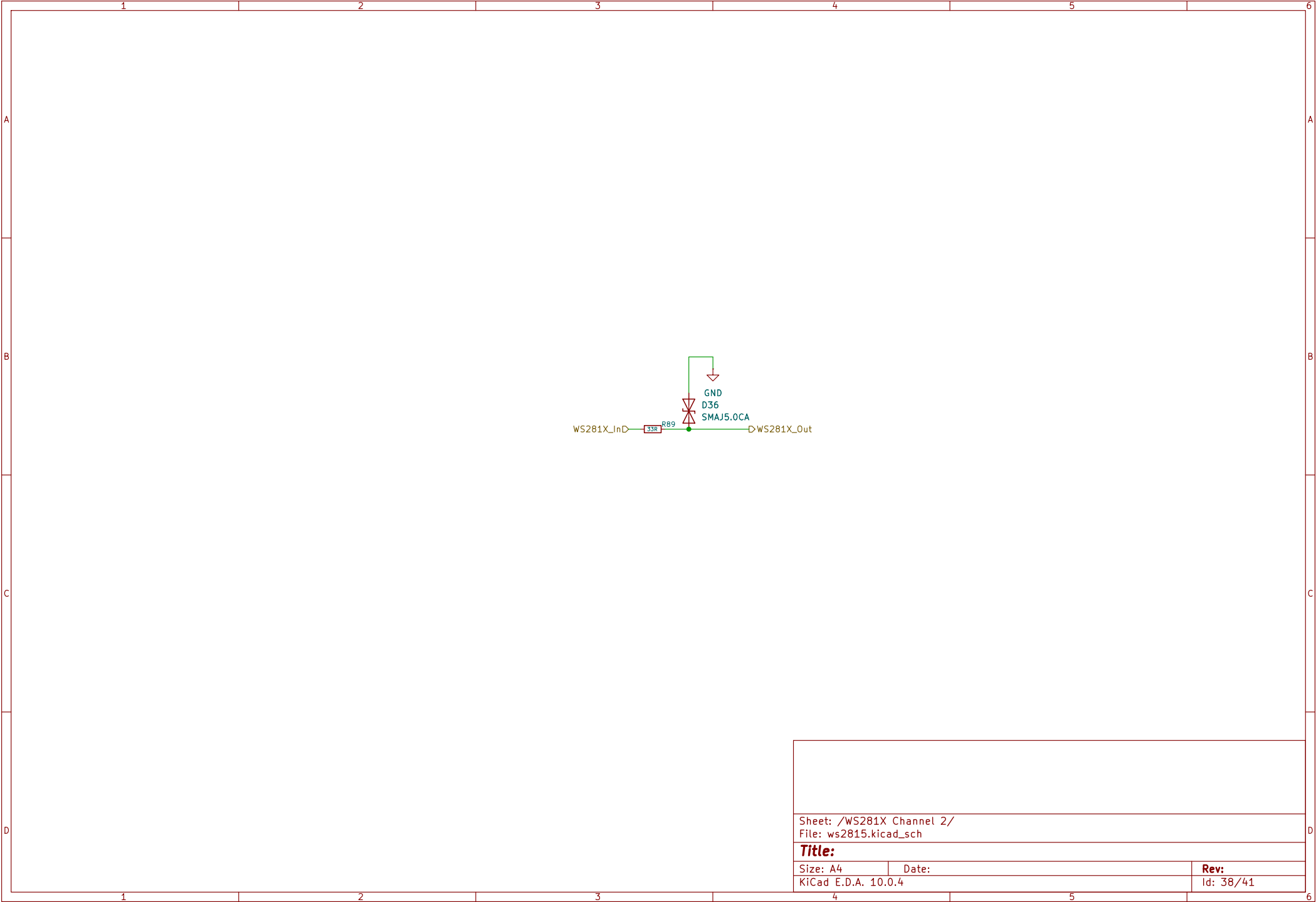
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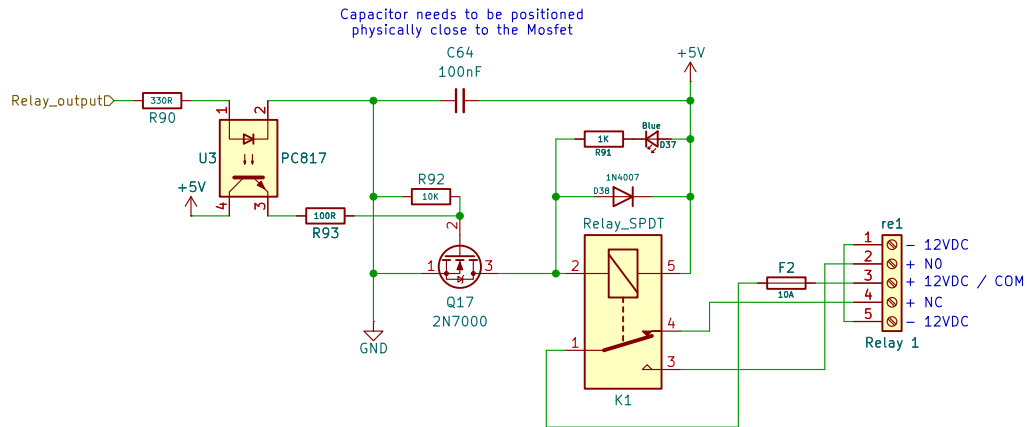
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Date:

Rev:
Id: 36/41







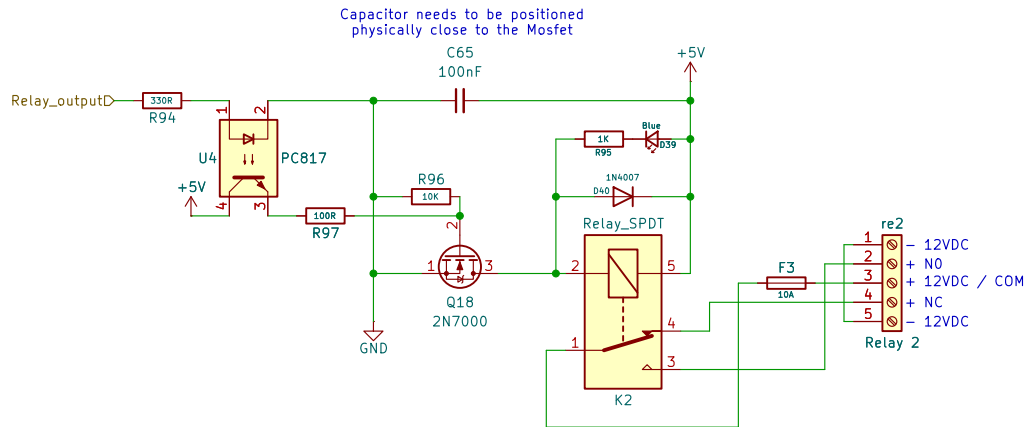
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Date:

Rev:
Id: 39/41



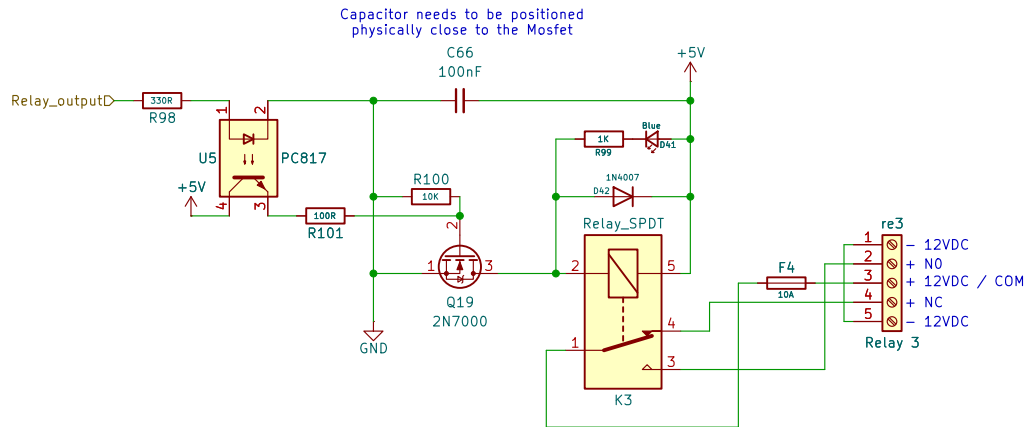
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Date:

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Id: 40/41



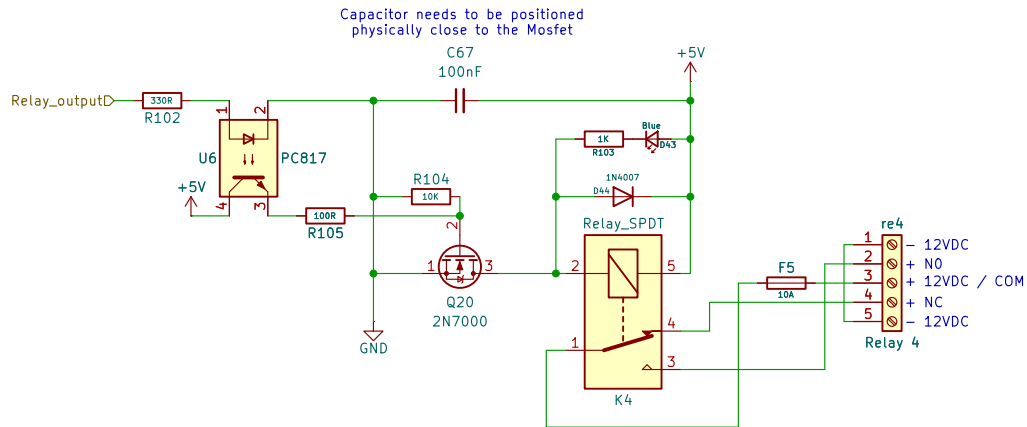
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KiCad E.D.A. 10.0.4

Date:

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Sheet: /Relay Channel 4/
File: relay_channel.kicad_sch

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Date:

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Id: 42/41